

Data Cleaning, Visualisation, String Manipulation

In []: IL EDA V3 H5 S3 CGX

```
In [1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# --- Step 1: Load the Dataset ---
# Load the Penguins dataset from seaborn
print("--- Loading Dataset ---")
penguins = sns.load_dataset("penguins")

# Display the first few rows
print("Original Dataset Head:")
print(penguins.head())

# --- Step 2: Exploratory Data Analysis (EDA) ---
print("\n--- Exploratory Data Analysis ---")

# Check dataset info
print("\nDataset Info:")
print(penguins.info())

# Describe numerical columns
print("\nStatistical Summary of Numerical Columns:")
print(penguins.describe())

# Check for missing values
print("\nMissing Data Count:")
print(penguins.isnull().sum())

# -- Visualizations --

# 1. Distribution of bill length
plt.figure(figsize=(8, 6))
sns.histplot(penguins['bill_length_mm'], kde=True, bins=20, color="blue")
plt.title("Distribution of Bill Length")
plt.xlabel("Bill Length (mm)")
plt.ylabel("Frequency")
plt.show()

# 2. Boxplot for flipper length by species
plt.figure(figsize=(8, 6))
# Note: Added hue=species to avoid FutureWarnings in newer Seaborn versions
sns.boxplot(x="species", y="flipper_length_mm", data=penguins, hue="species", palette="magma")
plt.title("Flipper Length by Species")
plt.xlabel("Species")
plt.ylabel("Flipper Length (mm)")
plt.show()
```

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# 3. Correlation heatmap
plt.figure(figsize=(10, 8))
# select_dtypes(include='number') is safer than 'float64' to include integers if an
sns.heatmap(penguins.select_dtypes(include='number').corr(), annot=True, cmap="cool")
plt.title("Correlation Heatmap")
plt.show()

# --- Step 3: Handle Missing Data ---
print("\n--- Handling Missing Data ---")

# Fill missing numerical values
# Note: We assign the result back to the column instead of using inplace=True
penguins["bill_length_mm"] = penguins["bill_length_mm"].fillna(penguins["bill_lengt
penguins["bill_depth_mm"] = penguins["bill_depth_mm"].fillna(penguins["bill_depth_m
penguins["flipper_length_mm"] = penguins["flipper_length_mm"].fillna(penguins["flip

# Fill body_mass_g with median (as per original code logic)
penguins["body_mass_g"] = penguins["body_mass_g"].fillna(penguins["body_mass_g"].me

# Fill missing categorical values with "Unknown"
penguins["sex"] = penguins["sex"].fillna("Unknown")

print("Missing Data Count After Cleaning:")
print(penguins.isnull().sum())

# --- Step 4: String Manipulation ---
print("\n--- String Manipulation ---")

# Create a new column by concatenating species and island
penguins["species_island"] = penguins["species"] + " - " + penguins["island"]

# Convert species names to uppercase
penguins["species"] = penguins["species"].str.upper()

# Extract the first three letters of the island name
penguins["island_code"] = penguins["island"].str[:3]

print("Dataset Head After Manipulation:")
print(penguins.head())

# --- Step 5: Save the Cleaned Dataset ---
cleaned_file_path = "cleaned_penguins.csv"
penguins.to_csv(cleaned_file_path, index=False)
print(f"\nCleaned dataset saved as CSV at: {cleaned_file_path}")

# --- Step 6: Grouped Analysis ---
print("\n--- Grouped Analysis ---")

# Group by species and calculate mean flipper length
grouped_data = penguins.groupby("species")["flipper_length_mm"].mean().reset_index()

print("Mean Flipper Length by Species:")
print(grouped_data)

# Save grouped data as Excel

```

```
grouped_file_path = "grouped_penguins.xlsx"
try:
    grouped_data.to_excel(grouped_file_path, index=False)
    print(f"Grouped data saved as Excel at: {grouped_file_path}")
except ImportError:
    print("\nError: Could not save Excel file. Please install openpyxl: 'pip instal
```

--- Loading Dataset ---

Original Dataset Head:

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	\
0	Adelie	Torgersen	39.1	18.7	181.0	
1	Adelie	Torgersen	39.5	17.4	186.0	
2	Adelie	Torgersen	40.3	18.0	195.0	
3	Adelie	Torgersen	NaN	NaN	NaN	
4	Adelie	Torgersen	36.7	19.3	193.0	

	body_mass_g	sex
0	3750.0	Male
1	3800.0	Female
2	3250.0	Female
3	NaN	NaN
4	3450.0	Female

--- Exploratory Data Analysis ---

Dataset Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 344 entries, 0 to 343

Data columns (total 7 columns):

#	Column	Non-Null Count	Dtype
0	species	344 non-null	object
1	island	344 non-null	object
2	bill_length_mm	342 non-null	float64
3	bill_depth_mm	342 non-null	float64
4	flipper_length_mm	342 non-null	float64
5	body_mass_g	342 non-null	float64
6	sex	333 non-null	object

dtypes: float64(4), object(3)

memory usage: 18.9+ KB

None

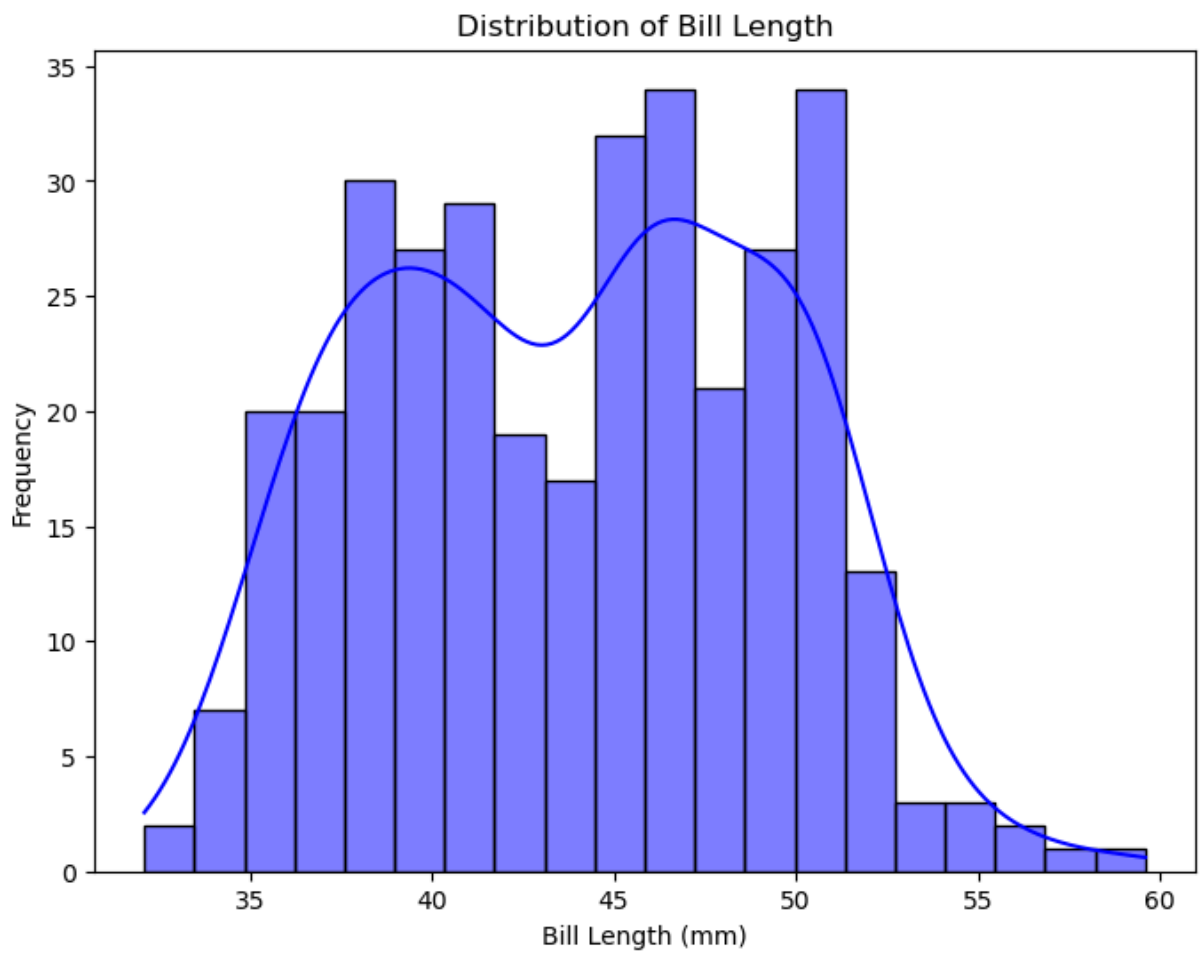
Statistical Summary of Numerical Columns:

	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
count	342.000000	342.000000	342.000000	342.000000
mean	43.921930	17.151170	200.915205	4201.754386
std	5.459584	1.974793	14.061714	801.954536
min	32.100000	13.100000	172.000000	2700.000000
25%	39.225000	15.600000	190.000000	3550.000000
50%	44.450000	17.300000	197.000000	4050.000000
75%	48.500000	18.700000	213.000000	4750.000000
max	59.600000	21.500000	231.000000	6300.000000

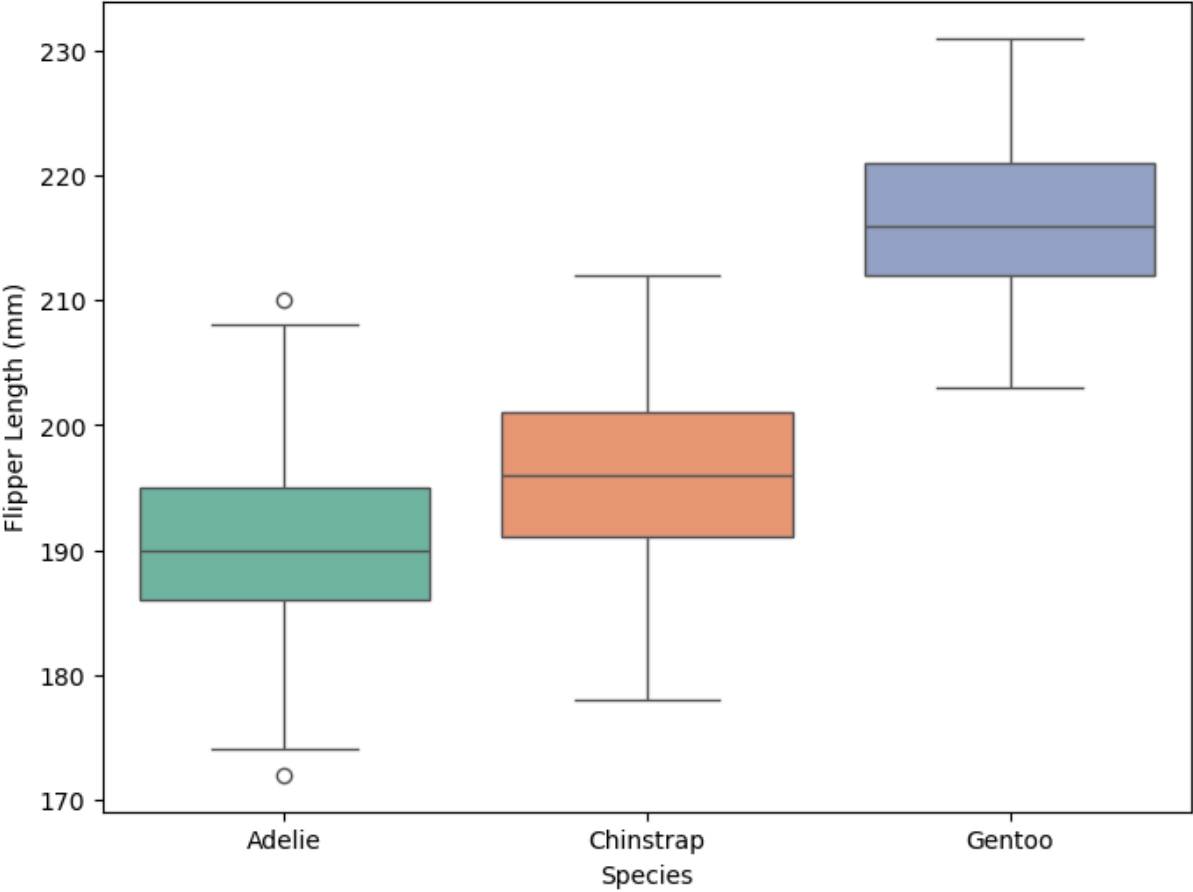
Missing Data Count:

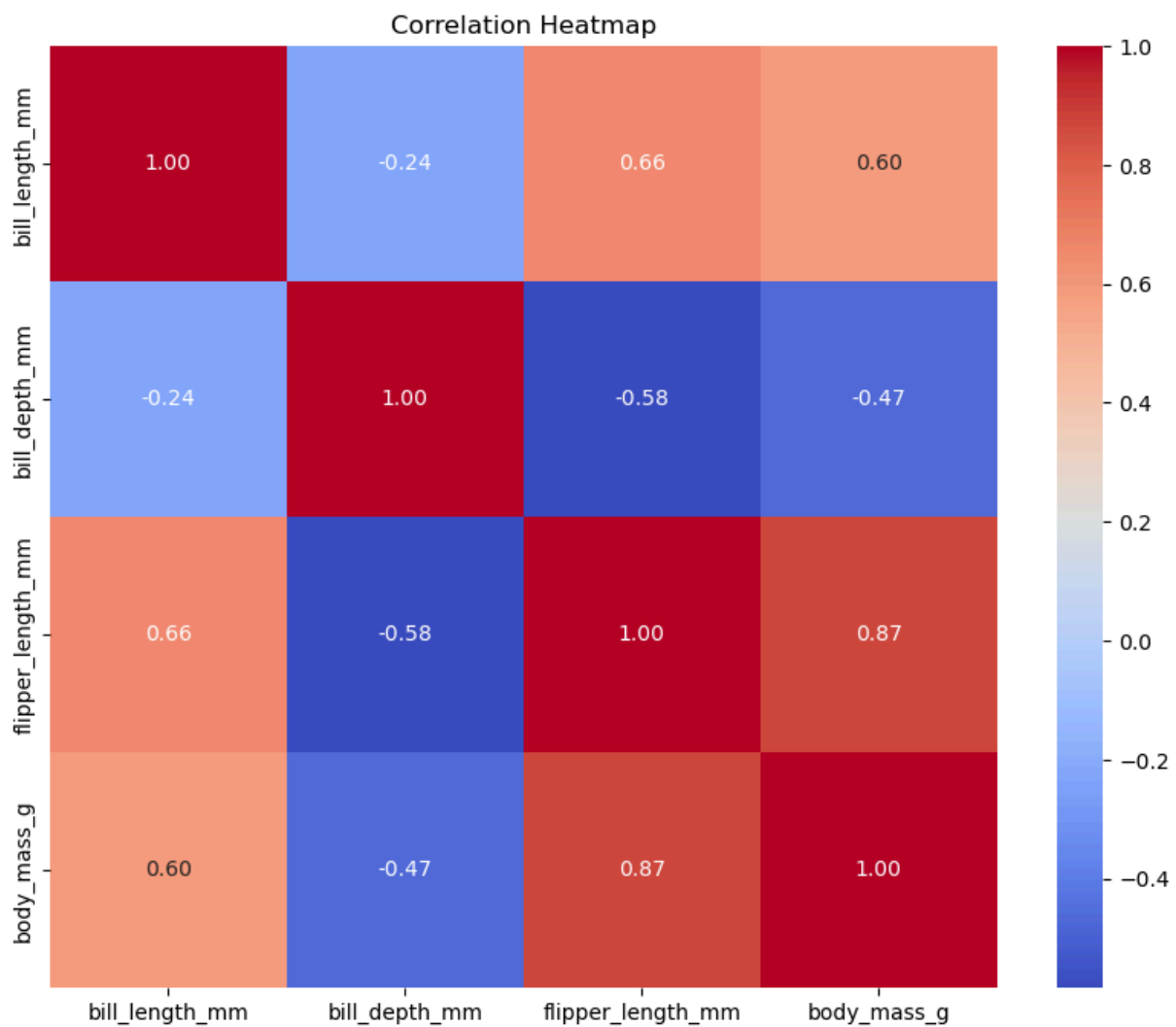
species	0
island	0
bill_length_mm	2
bill_depth_mm	2
flipper_length_mm	2
body_mass_g	2
sex	11

dtype: int64



Flipper Length by Species





--- Handling Missing Data ---

Missing Data Count After Cleaning:

```
species      0
island        0
bill_length_mm  0
bill_depth_mm  0
flipper_length_mm  0
body_mass_g    0
sex           0
```

dtype: int64

--- String Manipulation ---

Dataset Head After Manipulation:

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	\
0	ADELIE	Torgersen	39.10000	18.70000	181.000000	
1	ADELIE	Torgersen	39.50000	17.40000	186.000000	
2	ADELIE	Torgersen	40.30000	18.00000	195.000000	
3	ADELIE	Torgersen	43.92193	17.15117	200.915205	
4	ADELIE	Torgersen	36.70000	19.30000	193.000000	

	body_mass_g	sex	species_island	island_code
0	3750.0	Male	Adelie - Torgersen	Tor
1	3800.0	Female	Adelie - Torgersen	Tor
2	3250.0	Female	Adelie - Torgersen	Tor
3	4050.0	Unknown	Adelie - Torgersen	Tor
4	3450.0	Female	Adelie - Torgersen	Tor

Cleaned dataset saved as CSV at: cleaned_penguins.csv

--- Grouped Analysis ---

Mean Flipper Length by Species:

	species	flipper_length_mm
0	ADELIE	190.025758
1	CHINSTRAP	195.823529
2	GENTOO	217.055768

Grouped data saved as Excel at: grouped_penguins.xlsx

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

penguins = sns.load_dataset('penguins')
penguins.head()
```

```
Out[2]:
```

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	
0	Adelie	Torgersen	39.1	18.7	181.0	3750.0	I
1	Adelie	Torgersen	39.5	17.4	186.0	3800.0	Fei
2	Adelie	Torgersen	40.3	18.0	195.0	3250.0	Fei
3	Adelie	Torgersen	NaN	NaN	NaN	NaN	
4	Adelie	Torgersen	36.7	19.3	193.0	3450.0	Fei

```
In [5]: print(penguins.info())
print(penguins.describe())
print("Missing values in Dataset")
print(penguins.isnull().sum())
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 344 entries, 0 to 343
```

```
Data columns (total 7 columns):
```

#	Column	Non-Null Count	Dtype
0	species	344 non-null	object
1	island	344 non-null	object
2	bill_length_mm	342 non-null	float64
3	bill_depth_mm	342 non-null	float64
4	flipper_length_mm	342 non-null	float64
5	body_mass_g	342 non-null	float64
6	sex	333 non-null	object

```
dtypes: float64(4), object(3)
```

```
memory usage: 18.9+ KB
```

```
None
```

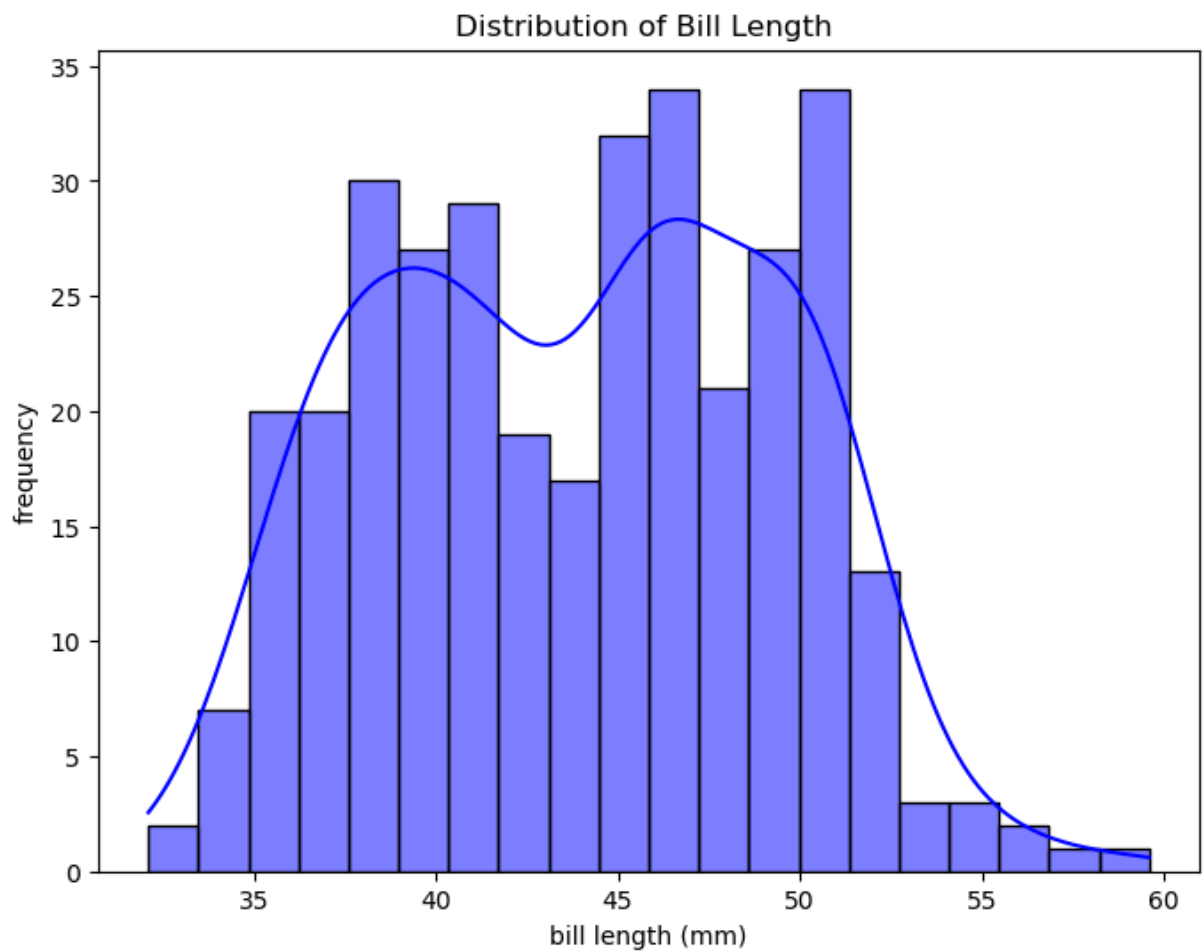
	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
count	342.000000	342.000000	342.000000	342.000000
mean	43.921930	17.151170	200.915205	4201.754386
std	5.459584	1.974793	14.061714	801.954536
min	32.100000	13.100000	172.000000	2700.000000
25%	39.225000	15.600000	190.000000	3550.000000
50%	44.450000	17.300000	197.000000	4050.000000
75%	48.500000	18.700000	213.000000	4750.000000
max	59.600000	21.500000	231.000000	6300.000000

```
Missing values in Dataset
```

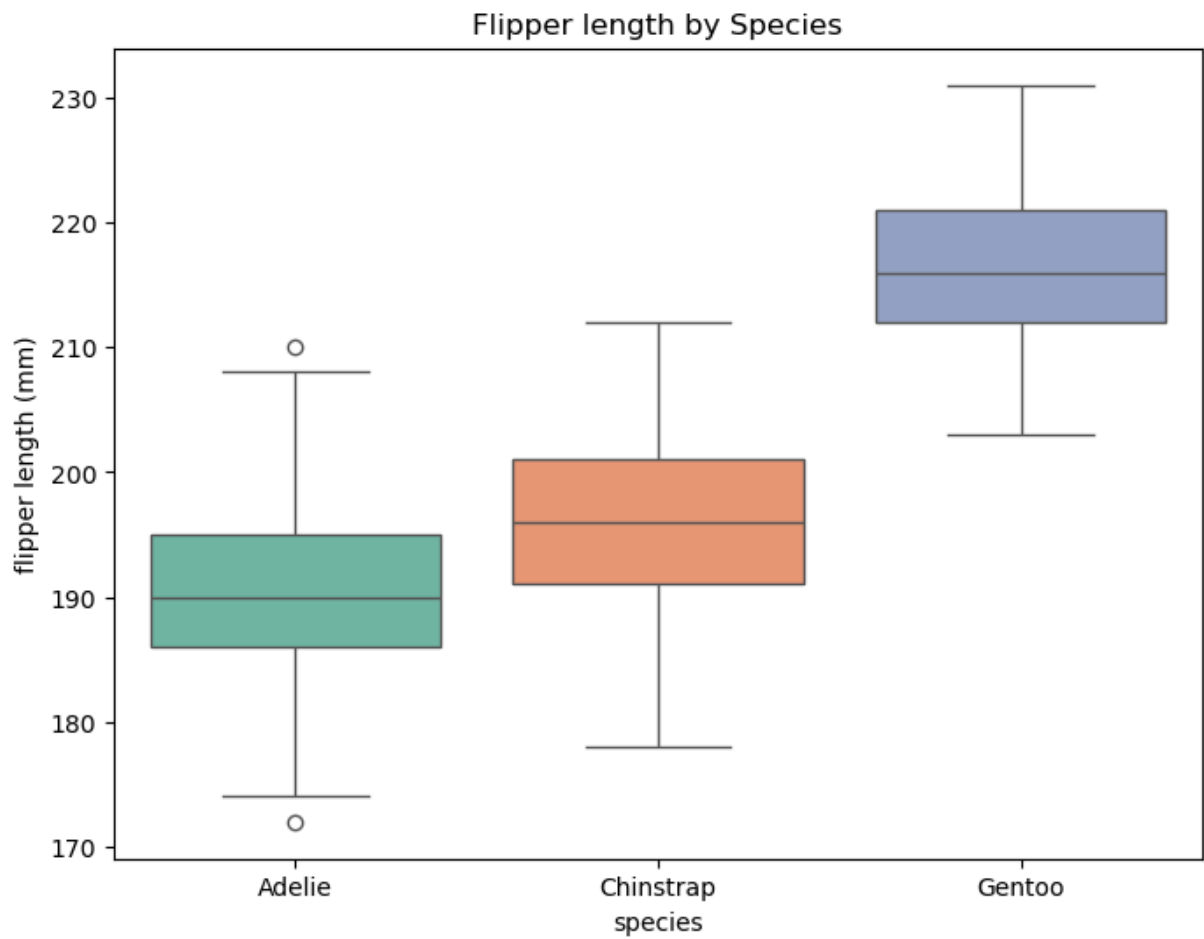
species	0
island	0
bill_length_mm	2
bill_depth_mm	2
flipper_length_mm	2
body_mass_g	2
sex	11

```
dtype: int64
```

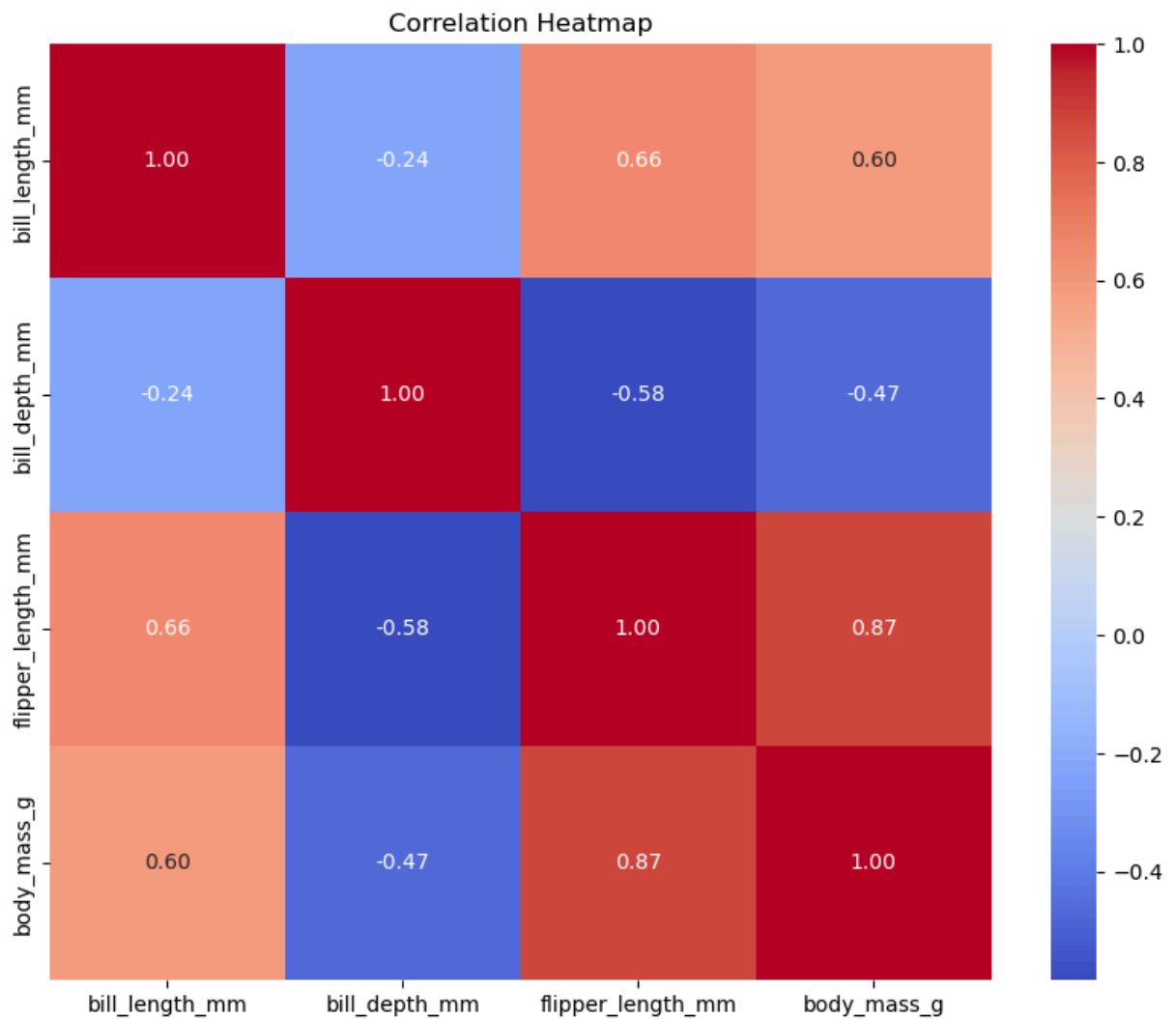
```
In [7]: plt.figure(figsize=(8,6))
sns.histplot(penguins['bill_length_mm'],kde=True,bins=20,color='blue')
plt.title("Distribution of Bill Length ")
plt.xlabel("bill length (mm)")
plt.ylabel("frequency")
plt.show()
```



```
In [8]: plt.figure(figsize=(8,6))
sns.boxplot(x='species',y='flipper_length_mm',data=penguins,hue='species',palette='
plt.title("Flipper length by Species")
plt.xlabel("species")
plt.ylabel("flipper length (mm)")
plt.show()
```



```
In [9]: plt.figure(figsize=(10,8))
sns.heatmap(penguins.select_dtypes(include='number').corr(),annot=True,cmap='coolwa
plt.title("Correlation Heatmap")
plt.show()
```



```
In [10]: penguins.head()
```

```
Out[10]:
```

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	
0	Adelie	Torgersen	39.1	18.7	181.0	3750.0	I
1	Adelie	Torgersen	39.5	17.4	186.0	3800.0	Fe
2	Adelie	Torgersen	40.3	18.0	195.0	3250.0	Fe
3	Adelie	Torgersen	NaN	NaN	NaN	NaN	
4	Adelie	Torgersen	36.7	19.3	193.0	3450.0	Fe

```
In [11]: penguins['bill_length_mm']=penguins['bill_length_mm'].fillna(penguins['bill_length_mm'].median())
penguins['bill_depth_mm']=penguins['bill_depth_mm'].fillna(penguins['bill_depth_mm'].median())
penguins['flipper_length_mm']=penguins['flipper_length_mm'].fillna(penguins['flipper_length_mm'].median())
penguins['body_mass_g']=penguins['body_mass_g'].fillna(penguins['body_mass_g'].median())
penguins['sex']=penguins['sex'].fillna('Unknown')
print(penguins.isnull().sum())
```

```

species      0
island       0
bill_length_mm 0
bill_depth_mm 0
flipper_length_mm 0
body_mass_g  0
sex          0
dtype: int64

```

```

In [13]: penguins['species_island'] = penguins['species']+"-"+penguins['island']
penguins['island_code'] = penguins["island"].str[:3]
penguins['species'] = penguins['species'].str.upper()
penguins.head()

```

Out[13]:

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
0	ADELIE	Torgersen	39.10000	18.70000	181.000000	3750.0
1	ADELIE	Torgersen	39.50000	17.40000	186.000000	3800.0
2	ADELIE	Torgersen	40.30000	18.00000	195.000000	3250.0
3	ADELIE	Torgersen	43.92193	17.15117	200.915205	4050.0
4	ADELIE	Torgersen	36.70000	19.30000	193.000000	3450.0

```

In [14]: csv_file_path = "cleaned_penguins.csv"
penguins.to_csv(csv_file_path,index=False)
print(f"Data saved at {csv_file_path}")

grouped_data = penguins.groupby("species")["flipper_length_mm"].mean().reset_index()
print(grouped_data)

```

```

Data saved at cleaned_penguins.csv
   species  flipper_length_mm
0  ADELIE          190.025758
1 CHINSTRAP          195.823529
2   GENTOO          217.055768

```

```

In [15]: excel_grouped_data = "grouped_data.xlsx"
penguins.to_excel(excel_grouped_data,index=False)
print(f"Data saved at {excel_grouped_data}")

```

```

Data saved at grouped_data.xlsx

```

```

In [ ]:

```